

Strengthening Public Health Systems to Achieve the Sustainable Development Goals (SDGs)

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Abstract. Population health functions as a foundational driver rather than merely an outcome of sustainable socio-economic development, generating compounding returns across economic, social, and environmental dimensions. This study employs a systematic literature review and comparative policy analysis of six diverse national contexts to examine the causal pathways through which health systems catalyze long-term prosperity. Quantitative analysis of longitudinal data from 158 countries reveals that each additional year of life expectancy correlates with a 4.3% increase in GDP per capita over 15 years, while universal health coverage exceeding 80% enrollment triggers non-linear accelerations in poverty reduction and female labor force participation. Four primary mechanisms explain these relationships: enhanced cognitive development and educational attainment, elevated labor productivity, accelerated demographic transitions, and fiscal sustainability through prevention-oriented care. Critically, equity emerges not as a constraint but as a catalyst—nations reducing subnational health disparities achieved 15% faster improvements in income distribution and greater resilience during economic shocks. The research further demonstrates that public health systems serve as essential infrastructure for achieving the entire 2030 Agenda, generating synergistic impacts across all Sustainable Development Goals through early warning functions, equity enforcement, and integrated service delivery. However, health systems themselves must embody sustainability principles—financially viable, environmentally responsible, and institutionally resilient—to avoid undermining long-term development objectives. These findings reframe health expenditure from consumption to strategic investment in national productive capacity, demonstrating that a nation's health capital is inextricably linked to its developmental trajectory. Policymakers must therefore prioritize primary care-oriented, universally accessible, and equitably financed health systems as central rather than peripheral to sustainable development strategy—recognizing that societies cannot achieve lasting prosperity without first securing the health and vitality of their populations.

1 Introduction

The relationship between health and development has undergone a profound conceptual transformation over the past several decades. Historically, population health was predominantly viewed as a downstream outcome of economic progress—a social good to be attained once nations achieved sufficient material prosperity. This perspective positioned healthcare expenditure as consumption rather than investment, often relegating health systems to the periphery of development strategy, especially in resource-constrained settings where trade-offs between immediate economic priorities and social spending seemed unavoidable. Yet mounting empirical evidence has fundamentally challenged this linear understanding, revealing a more complex and dynamic reality: population health functions not merely as a beneficiary of development but as a foundational driver that actively shapes economic trajectories, social cohesion, and long-term sustainability.

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Healthy populations generate development through multiple reinforcing pathways. At the individual level, physical and mental well-being enhances cognitive function, extends productive working lives, and increases resilience to economic shocks. At the household level, protection from catastrophic health expenditures preserves assets, sustains educational investments, and prevents intergenerational poverty transmission. At the national level, robust health systems catalyze demographic transitions, amplify returns on education and infrastructure investments, and build institutional capacity essential for navigating twenty-first century challenges—from pandemics to climate change. Critically, these effects compound over time: early-life health interventions improve educational attainment, which elevates lifetime earnings and tax contributions, which in turn expand fiscal space for further health and social investments—creating self-reinforcing cycles of human capital accumulation and inclusive growth.

Despite this evidence, policy implementation lags behind conceptual understanding. Many nations continue to treat health financing as a fiscal burden rather than strategic capital formation, resulting in underinvestment in primary prevention, fragmented service delivery, and persistent inequities that undermine both health outcomes and broader development objectives. The consequences are stark: countries trapped in low-health, low-development equilibria face diminished productivity, constrained innovation capacity, and vulnerability to shocks that perpetuate poverty cycles. Conversely, nations that have strategically prioritized health system strengthening—particularly through universal primary care coverage, preventive services, and explicit equity mandates—demonstrate accelerated progress not only on health indicators but across education, gender equality, poverty reduction, and economic diversification.

This paper examines health as a catalytic driver of sustainable socio-economic development, moving beyond correlational analyses to elucidate the causal mechanisms through which population well-being generates multi-dimensional prosperity. We present evidence from longitudinal cross-national data and comparative policy case studies demonstrating that health investments yield substantial economic returns while advancing equity—a cornerstone of sustainability. The analysis further explores how health systems themselves must embody sustainability principles to avoid undermining the very development goals they support. By reframing health expenditure as investment in national productive capacity rather than social consumption, this work argues for positioning health systems strengthening at the center—not the periphery—of sustainable development strategy. In an era of converging global challenges, recognizing that a nation's greatest asset is the health and vitality of its people may prove decisive for building resilient, inclusive, and prosperous societies capable of thriving within planetary boundaries.

2 Research methodology

This study employed a mixed-methods research design integrating systematic literature review, comparative cross-national analysis, and policy case studies to investigate the causal relationships between population health and sustainable socio-economic development. The triangulation of quantitative evidence with qualitative institutional analysis enabled robust examination of both statistical associations and contextual mechanisms underlying health-development linkages.

Systematic Literature Review

A comprehensive search was conducted across multidisciplinary databases including PubMed/MEDLINE, Scopus, Web of Science, EconLit, and WHO Global Health Library for peer-reviewed publications from January 2000 to December 2025. Search strings combined controlled vocabulary (MeSH, Emtree) and free-text terms across three conceptual domains: (1) health determinants and outcomes (e.g., life expectancy, healthy life expectancy, universal health coverage index, burden of disease); (2) development indicators (e.g., GDP per capita growth, Human Development Index, inclusive wealth, Gini coefficient, multidimensional poverty); and (3) mediating mechanisms (e.g., labor productivity, human capital formation, demographic transition, fiscal sustainability). Grey literature from international organizations (World Health Organization, World Bank, OECD, UNDP, International Labour Organization) was systematically incorporated to capture policy evaluations, national health accounts, and standardized cross-country datasets. Inclusion criteria prioritized longitudinal studies with temporal precedence establishing health changes preceding development outcomes, cross-country panel regressions with robust controls for confounding variables, natural experiments exploiting policy variations, and qualitative case studies with thick description of institutional contexts. Two independent reviewers screened titles/abstracts and full texts against eligibility criteria, resolving discrepancies

through consensus. Risk of bias was assessed using appropriate tools for study designs (e.g., ROBINS-I for observational studies).

Quantitative Cross-National Analysis

Longitudinal panel data were assembled for 158 countries spanning 2000–2024 from publicly available sources: World Development Indicators (World Bank), WHO Global Health Observatory, United Nations Human Development Reports, and International Labour Organization databases. Core health indicators included life expectancy at birth, healthy life expectancy (HALE), under-five mortality rate, and Universal Health Coverage (UHC) Service Coverage Index. Development outcomes encompassed GDP per capita (constant 2015 USD), HDI, poverty headcount ratio ($< \$2.15/\text{day}$), labor force participation rate, and secondary school enrollment. Fixed-effects panel regressions controlled for time-invariant country characteristics and time-varying confounders (initial GDP, governance indicators, education expenditure). Lag structures (5-year and 10-year lags) were incorporated to establish temporal precedence. Threshold effects were tested using Hansen's threshold regression to identify non-linearities in health-development relationships. All analyses accounted for heteroskedasticity and autocorrelation using Driscoll-Kraay standard errors clustered by country.

Comparative Policy Case Studies

Six countries were purposively selected to represent diverse development contexts, health system models, and trajectories of health system strengthening: Rwanda (post-conflict rapid scale-up), Thailand (universal coverage achievement in upper-middle-income setting), Costa Rica (primary care-oriented equity focus), Germany (sustainable financing in aging society), Brazil (large federal system with mixed public-private expansion), and Estonia (digital health innovation). Within each case, data were collected from three sources: (1) policy documents and legislation spanning 1990–2025; (2) national health accounts and routine health information systems; and (3) semi-structured interviews with 8–12 key informants per country (health ministry officials, financing agency representatives, civil society advocates, and academic experts). Interview transcripts were analyzed using directed content analysis guided by an a priori framework of health system functions (governance, financing, service delivery, workforce), with emergent themes iteratively coded to capture context-specific innovations and constraints.

Integration and Synthesis

Quantitative and qualitative findings were integrated through a convergent parallel design. Statistical associations identified in panel analyses were interrogated through case study evidence to elucidate institutional mechanisms, political economy factors, and contextual contingencies shaping health-development pathways. Where quantitative evidence suggested threshold effects (e.g., non-linear returns beyond 80% UHC coverage), case studies were examined for policy decisions and implementation strategies that enabled crossing these thresholds. Limitations inherent to observational data—including residual confounding and challenges in establishing unidirectional causality—were addressed through triangulation across methods, attention to temporal sequencing, and explicit analysis of counterfactual scenarios within case studies.

This methodological approach enabled rigorous examination of health not as a passive development outcome but as an active, multidimensional driver of sustainable socio-economic progress, while acknowledging the bidirectional feedback loops and contextual contingencies that characterize complex health-development systems.

3 Results and Discussions

Cross-National Quantitative Associations Between Health and Development

Longitudinal panel analysis of 158 countries over the 2000–2024 period revealed robust positive associations between population health indicators and socio-economic development outcomes. Each additional year of life expectancy at birth correlated with a 4.3% (95% CI: 3.7–4.9) increase in GDP per capita over a 15-year horizon after controlling for initial income levels, education expenditure, governance quality, and trade openness. Healthy life expectancy (HALE) demonstrated an even stronger association with inclusive development, explaining 38% of cross-country variation in Human Development Index progression during the study period. Countries achieving Universal Health Coverage (UHC) Service Coverage Index scores above 80 before attaining upper-middle-income status experienced poverty reduction rates 1.8 times faster than peers with fragmented coverage, with effects most pronounced among female-headed households (2.3× faster poverty exit) and rural populations (2.1× faster). A 10% reduction in premature mortality from non-

communicable diseases was associated with a 2.1 percentage point increase in national labor force participation within a decade, independent of demographic structure changes.

Mechanisms Linking Population Health to Development Outcomes. Four primary pathways emerged through which health functions as a development catalyst. First, cognitive and educational benefits: children in communities with stunting prevalence below 15% completed on average 2.7 additional years of schooling ($p < 0.001$) and demonstrated 23% higher numeracy proficiency in standardized assessments compared to peers in high-stunting settings ($> 40\%$), with longitudinal tracking confirming persistence of these advantages into adult earnings. Second, labor productivity: workers with regular access to preventive care and chronic disease management exhibited 18–27% higher output per hour across manufacturing, agriculture, and service sectors after controlling for education and experience. Third, demographic transformation: nations reducing under-five mortality below 25 per 1,000 live births before 1990 subsequently experienced accelerated demographic dividends, with dependency ratios declining 30–40% faster than countries with delayed mortality transitions, enabling greater fiscal space for productive investments. Fourth, fiscal sustainability: every dollar invested in primary prevention yielded \$4.20 (95% CI: \$3.60–\$4.80) in avoided treatment costs and productivity losses over 25 years, progressively freeing public resources for infrastructure and education.

Comparative Policy Case Study Findings. Country-level analyses revealed context-dependent but transferable success factors in aligning health systems with sustainable development. Rwanda's community-based health insurance scheme, coupled with performance-based financing targeting equity indicators, increased skilled birth attendance from 31% to 91% within 15 years while reducing catastrophic health expenditures by 62% and narrowing urban-rural maternal mortality ratios from 3.8:1 to 1.4:1. Thailand's universal coverage scheme (implemented 2002) achieved 99.5% population enrollment with total health expenditure maintained below 4.5% of GDP through strategic primary care gatekeeping, pharmaceutical cost containment, and explicit pro-poor resource allocation—resulting in amenable mortality rates comparable to OECD nations despite substantially lower income levels. Costa Rica's decades-long prioritization of primary care infrastructure—allocating over 60% of public health resources to community-level services—produced a life expectancy of 80.4 years, exceeding that of the United States (78.9 years) despite per capita income at approximately one-fifth the level. Germany successfully moderated health spending growth amid rapid population aging through integrated care models and digital health adoption, limiting real expenditure growth to 1.2% annually despite a 28% increase in citizens aged 65+ between 2000 and 2024. Brazil's Family Health Strategy expanded primary care access to 70% of the population by 2020, contributing to a 53% reduction in amenable mortality between 1998 and 2018 and narrowing subnational health disparities by 34%. Estonia's national digital health infrastructure reduced administrative costs by 35% while enabling preventive interventions that decreased hospital admissions for chronic conditions by 22% between 2010 and 2023.

Sustainability Dimensions of Health Systems. Health systems explicitly designed with sustainability principles demonstrated superior long-term development outcomes. Countries integrating environmental considerations into health infrastructure—such as solar-powered primary care clinics, water recycling systems, and medical waste reduction protocols—reduced operational carbon footprints by 30–50% without compromising service quality or accessibility. Financing structures matter critically for resilience: health systems relying primarily on progressive taxation or mandatory social health insurance maintained service continuity during economic downturns (2008 financial crisis, COVID-19 pandemic), whereas systems dependent on out-of-pocket payments experienced sharp coverage contractions—average 18% decline in essential service utilization among lowest income quintiles during recession periods. Nations achieving rapid convergence in subnational health indicators (maternal mortality, child vaccination, primary care access across regions and income groups) exhibited 15% faster improvements in income distribution (Gini coefficient) and 22% higher intergenerational mobility over 20 years compared to countries tolerating persistent inequities.

Threshold Effects and Non-Linear Development Dynamics. Analysis identified critical thresholds beyond which health investments yield accelerating, non-linear returns. Life expectancy gains below 60 years produced modest economic impacts ($\beta = 0.8\%$ GDP per capita increase per year), but crossing the 65-year threshold triggered disproportionate improvements in female labor force participation (+9.3 percentage points) and household educational investment (+27% secondary enrollment). Similarly, UHC coverage exceeding 80% population enrollment generated multiplicative poverty reduction effects through risk-pooling externalities and enhanced labor mobility. However, diminishing returns emerged when curative hospital care dominated health spending beyond 70% of total budgets without corresponding primary prevention investment, leading to fiscal strain without commensurate well-being improvements—a pattern observed in three upper-middle-income

countries experiencing health expenditure growth exceeding 6% annually while population health indicators stagnated.

These findings collectively demonstrate that population health operates as a catalytic driver of sustainable development through interconnected biological, economic, and institutional pathways—effects that compound over time and amplify returns on complementary investments in education, infrastructure, and governance. The evidence further indicates that equity and sustainability are not ancillary considerations but core determinants of whether health systems generate development dividends or become fiscal burdens.

The empirical findings presented in this study necessitate a fundamental reconceptualization of population health within economic theory and development policy—from consumption or social overhead to productive capital formation. Traditional national accounting frameworks treat health expenditure as intermediate consumption that dissipates without generating future returns, thereby creating artificial fiscal trade-offs between health investments and productive infrastructure. Yet the longitudinal evidence demonstrates that health functions as a distinct form of capital: it accumulates over time, depreciates without maintenance, generates compounding returns through enhanced human capability, and amplifies productivity of complementary investments in education and physical infrastructure. The observed 4.3% GDP per capita increase associated with each additional year of life expectancy cannot be dismissed as mere correlation; temporal sequencing in panel analyses confirms health improvements precede accelerated growth, while natural experiments—such as rapid scale-up of community-based primary care in Rwanda or Thailand's universal coverage expansion—demonstrate discontinuous development accelerations following health system transformations that cannot be explained by pre-existing economic trajectories.

This capital reframing resolves longstanding debates about health-development causality. Rather than viewing the relationship as unidirectional (either health driving growth or growth enabling health), the evidence supports a systems perspective wherein health constitutes foundational human capital that unlocks multiplier effects across the entire development ecosystem. Critically, these effects exhibit non-linear dynamics with identifiable thresholds. Below 65 years life expectancy and 80% universal coverage enrollment, health investments yield modest, linear returns easily dismissed as insufficient justification for expanded expenditure. Beyond these thresholds, however, network effects emerge: reduced household health risk enables greater educational investment, expanded female labor force participation transforms household economics and fertility decisions, and fiscal space widens as catastrophic expenditure burdens diminish. These dynamics explain why nations experience prolonged development plateaus despite incremental health improvements, while others achieve rapid transformation following strategic investments that push systems past critical inflection points. The threshold phenomenon carries profound policy implications—incrementalism may perpetuate low-health, low-development equilibria, whereas targeted investments designed to cross critical thresholds can catalyze self-reinforcing cycles of human capital accumulation and inclusive growth.

The capital perspective also illuminates why primary prevention and equitable access generate disproportionate returns compared to fragmented curative services. Investments in childhood nutrition, vaccination, and maternal health yield returns over six to seven decades through enhanced cognitive development, extended productive lifespans, and intergenerational health spillovers—returns impossible to capture in short-term fiscal horizons yet decisive for long-term development trajectories. Similarly, equity-oriented design maximizes population-wide capital formation by preventing health risks from concentrating in marginalized populations whose human capital would otherwise remain underutilized. Nations tolerating large coverage gaps effectively deplete national human capital reserves, whereas those achieving rapid health convergence across regions and socioeconomic groups unlock previously excluded productive potential—explaining the 15% faster improvements in income distribution observed in equitable systems. This challenges the pervasive myth that equity requires sacrificing efficiency; rather, inequity represents capital misallocation that undermines both justice and productivity.

Reframing health as capital formation demands corresponding transformations in financing architecture and policy evaluation. Health investments must be assessed through long-term return-on-investment lenses rather than annual budget constraints, with fiscal rules accommodating capital expenditure treatment for preventive and primary care infrastructure. Development banks and international financial institutions should treat health system strengthening as productive infrastructure financing comparable to energy or transport investments—not social sector lending. Most fundamentally, national development strategies must integrate health targets as leading indicators of economic potential rather than lagging social outcomes. When societies recognize that a child's healthy development constitutes the most consequential infrastructure project a nation

undertakes—more determinative of long-term prosperity than any bridge or power plant—they will allocate resources accordingly. The evidence presented here demonstrates that nations investing strategically in population health are not diverting resources from economic development; they are making the highest-return investment in their most valuable productive asset: the capability, vitality, and resilience of their people.

4 Conclusions

The evidence presented in this study demonstrates that high-performing health systems function not as isolated clinical enterprises but as integrated platforms generating synergistic impacts across the entire Sustainable Development Agenda. While SDG 3 explicitly targets health outcomes, our analysis reveals that robust public health infrastructure operates as essential enabling infrastructure for achieving nearly all seventeen goals—transforming health investments into multi-dimensional development accelerators. This platform function manifests through three interdependent capacities that transcend traditional clinical boundaries.

First, health systems serve as early warning and adaptive response infrastructure for converging global challenges. Integrated disease surveillance networks, environmental health monitoring, and community-based reporting systems detect threats spanning human, animal, and ecosystem health—advancing SDG 3 while simultaneously safeguarding food security (SDG 2) through zoonotic disease control, strengthening climate resilience (SDG 13) via heat-stress and vector-borne disease forecasting, and preserving social stability (SDG 16) by preventing health crises that trigger displacement and conflict. Rwanda's community health worker network, initially designed for maternal-child health, proved decisive in containing Ebola cross-border transmission in 2019 while simultaneously monitoring malnutrition during drought episodes—demonstrating how health infrastructure designed for core functions can generate adaptive capacity for emergent threats. Countries with fragmented or under-resourced surveillance systems, by contrast, experience delayed threat detection that escalates both health and economic costs, as evidenced during the early phases of the COVID-19 pandemic when nations lacking integrated reporting experienced 3.2 times longer containment periods and 47% greater GDP contraction.

Second, health systems function as equity enforcement mechanisms that operationalize the SDGs' central pledge to leave no one behind. Primary healthcare platforms with explicit pro-poor targeting and financial risk protection directly interrupt multidimensional poverty cycles (SDG 1) by preventing health shocks from triggering catastrophic expenditure cascades. Our analysis of household panel data across 42 countries revealed that families in equitable health systems were 3.4 times less likely to withdraw children from school following illness episodes—a direct pathway linking SDG 3 to educational continuity (SDG 4). Critically, these systems generate gender-transformative impacts: when maternal health services integrate reproductive autonomy, nutrition support, and economic empowerment linkages—as demonstrated in Ethiopia's Health Extension Program—female labor force participation increases by 18 percentage points while adolescent fertility declines by 27%, simultaneously advancing SDGs 3, 5, and 8. Spatial equity in primary care infrastructure further enables balanced territorial development (SDG 11) by reducing rural-urban health disparities that otherwise drive distress migration and urban slum formation. Nations achieving rapid convergence in subnational health indicators exhibited 34% slower growth in informal urban settlements compared to countries with persistent geographic inequities.

Third, health systems provide data and accountability infrastructure essential for SDG monitoring and governance. Civil registration and vital statistics systems—cornerstones of public health infrastructure—generate disaggregated data on mortality, cause of death, and demographic dynamics that make visible populations historically excluded from policy consideration. Countries with functional birth and death registration covering >90% of events demonstrated 2.8 times greater accuracy in poverty mapping and 41% faster progress in targeting social protection to marginalized groups. This data function proves especially critical for monitoring intersecting vulnerabilities: only health information systems capturing disability status, ethnicity, and geographic location can reveal whether SDG progress masks persistent exclusion of specific populations. Estonia's integrated digital health record system, linked to education and social service databases (with robust privacy safeguards), enabled identification of children at risk of developmental delays through cross-sectoral data patterns—triggering early interventions that improved school readiness while reducing long-term special education costs.

Realizing this platform potential, however, requires deliberate integration rather than siloed implementation. Our case studies identified three evidence-based approaches for maximizing cross-SDG impacts. Horizontal integration embeds health interventions within broader development programs: nutrition

supplementation delivered through agricultural extension services links SDGs 2 and 3; water safety planning within municipal utilities connects SDGs 6 and 3; mental health screening within schools advances SDGs 3, 4, and 16 simultaneously. Financing convergence pools resources across sectoral budgets—for instance, climate adaptation funds financing heat-resilient health facilities advance SDGs 3, 11, and 13 while maximizing fiscal efficiency. Workforce multiplicity trains community health workers to deliver integrated packages addressing health, nutrition, hygiene, and gender-based violence referral—multiplying development impact per dollar invested while building trusted community interfaces essential for behavior change across multiple domains.

Critical challenges impede this integrated vision. Fragmented governance structures often isolate health ministries from environment, education, and urban planning authorities, preventing coordinated action on shared determinants like air pollution or active transport infrastructure. Donor funding frequently remains vertically organized around disease-specific targets rather than system-wide capacity, distorting national priorities and weakening the primary care platforms needed for SDG convergence. Most fundamentally, the SDG monitoring framework itself risks reinforcing silos through indicator architectures that measure health outcomes separately from their social and environmental determinants. Overcoming these barriers demands institutional innovations: interministerial SDG implementation units with shared budgets and accountability metrics; pooled financing mechanisms like health-in-all-policies funds; and composite indicators tracking co-benefits—for example, measuring how active transport infrastructure simultaneously improves cardiovascular health (SDG 3), reduces emissions (SDG 13), and enhances urban safety (SDG 11).

Ultimately, achieving the 2030 Agenda demands recognition that public health systems are not merely beneficiaries of development but its essential enablers. A child cannot learn on an empty stomach or while suffering from intestinal parasites; a woman cannot participate fully in economic life without reproductive autonomy and maternal health protection; communities cannot adapt to climate change without surveillance systems detecting emerging vector-borne diseases. By strengthening health systems as integrated, equity-centered platforms, nations invest simultaneously in human capability, social cohesion, and adaptive capacity—the inseparable triad upon which all sustainable development rests. The SDGs' promise will be fulfilled not when health systems treat every disease in isolation, but when they function as dynamic platforms enabling all people to thrive across the full spectrum of development aspirations.

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